AI Engineering Concepts

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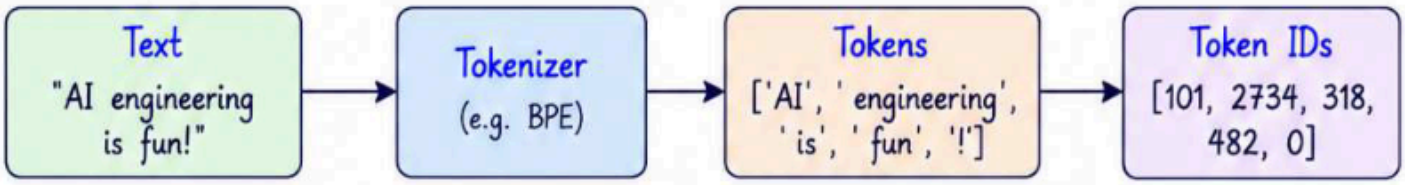
Key Ideas Behind Modern AI Systems

1 Tokenization

Text → Tokens (model readable pieces)

- Tokenization converts text into smaller pieces called tokens.
- These tokens can be words, subwords or even characters, depending on the tokenizer.

How it works?



Key Points

- Tokens are not always full words.
- Different models use different tokenizers.
- Tokenization helps convert text into numeric IDs.

Use Cases

- Preparing input for LLMs
- Counting tokens (for cost/limits)
- Text preprocessing

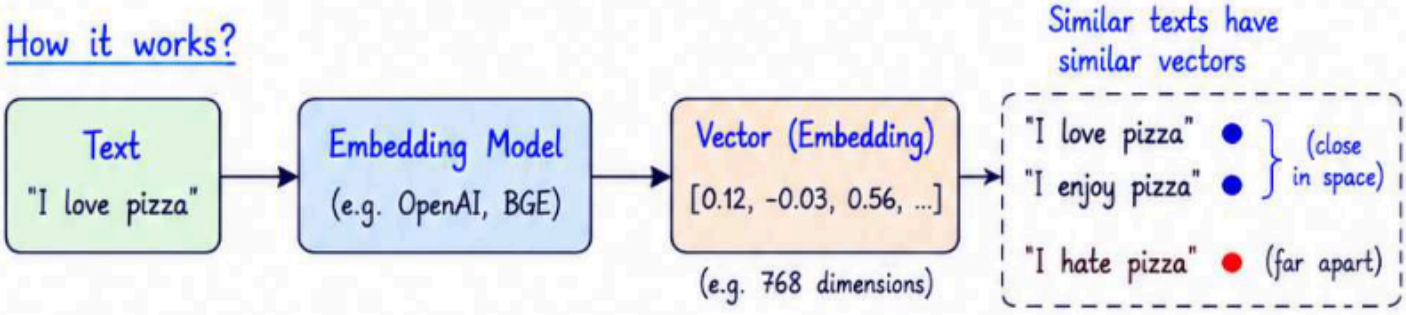
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2 Embeddings

Text → Vectors (capture meaning)

- Embeddings convert text into dense numeric vectors that capture the semantic meaning.
- Similar text will have similar embeddings even if the words are different.

How it works?



Key Points

- Embeddings capture meaning, not exact words.
- Similar content → similar vectors.
- Common dimension sizes: 384, 768, 1536 etc.

Use Cases

- Semantic search (e.g, in RAG)
- Finding similar documents
- Clustering content

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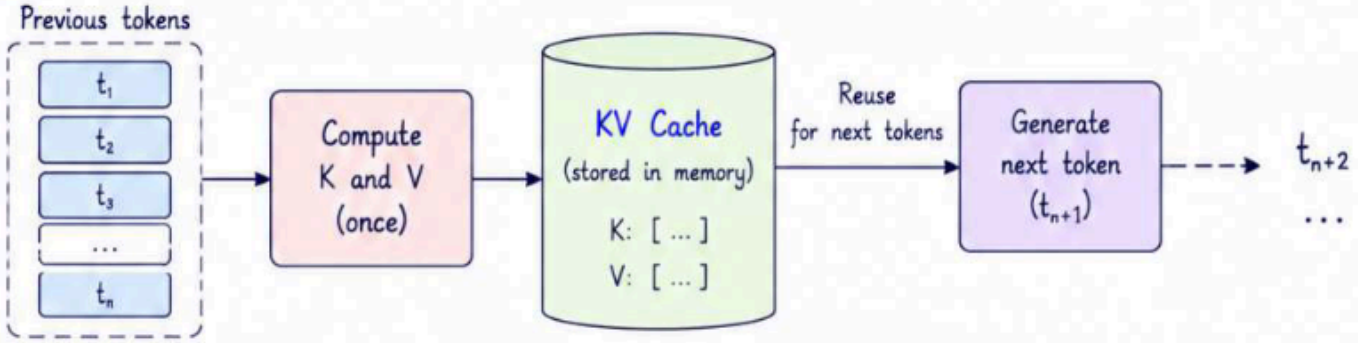
Key Ideas Behind Modern AI Systems

5 KV Cache

Reuse past computation

- KV Cache stores the key and value representations of previously processed tokens.
- It avoids recomputing them for every new token, making generation faster and more efficient.

How it works?



Key Points

- Caches key and value vectors of past tokens.
- Reduces redundant computation.
- Significantly improves generation speed.

Use Cases

- Faster text generation (LLMs)
- Real-time chat applications
- Serving LLMs efficiently

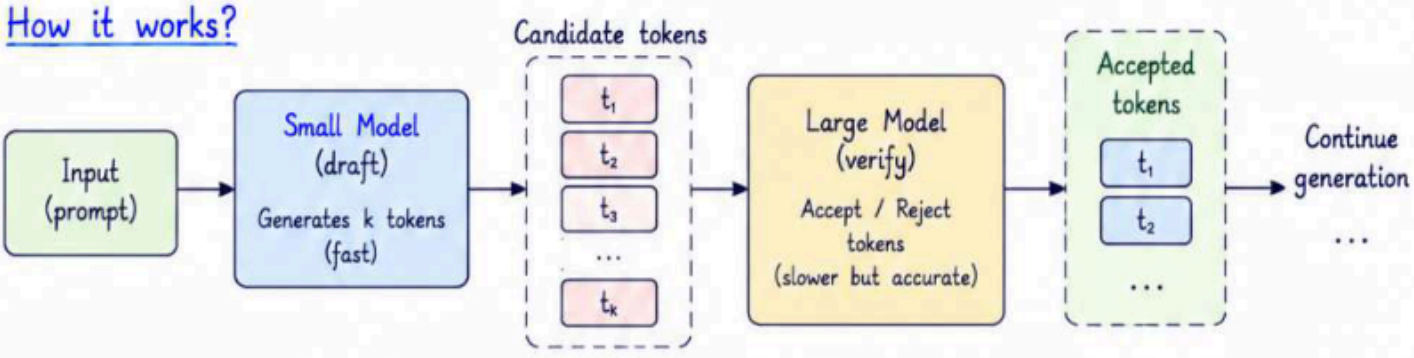
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6 Speculative Decoding

Draft fast, verify with a stronger model

- A smaller, faster model first generates multiple candidate tokens.
- A larger, more accurate model then verifies them, accepting correct tokens and rejecting the rest.

How it works?



Key Points

- Uses a small model to draft tokens.
- Large model verifies the drafted tokens.
- Improves generation speed without losing quality.

Use Cases

- Faster inference for LLMs
- Reducing latency in real-time applications
- Efficient serving at scale

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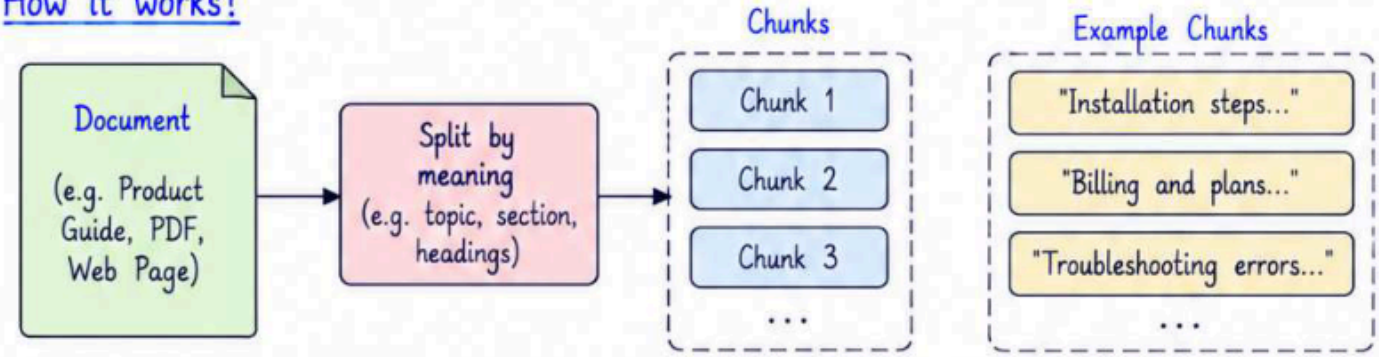
Key Ideas Behind Modern AI Systems

7 Semantic Chunking

Split documents by meaning

- Semantic chunking divides a document into smaller chunks based on meaning, not just fixed size.
- This creates more relevant chunks for retrieval and reduces loss of context.

How it works?



Key Points

- Splits text based on semantic meaning.
- Produces more relevant and coherent chunks.
- Improves retrieval quality in RAG systems.

Use Cases

- Preparing documents for RAG
- Indexing long documents (PDFs, manuals)
- Better retrieval and answer quality

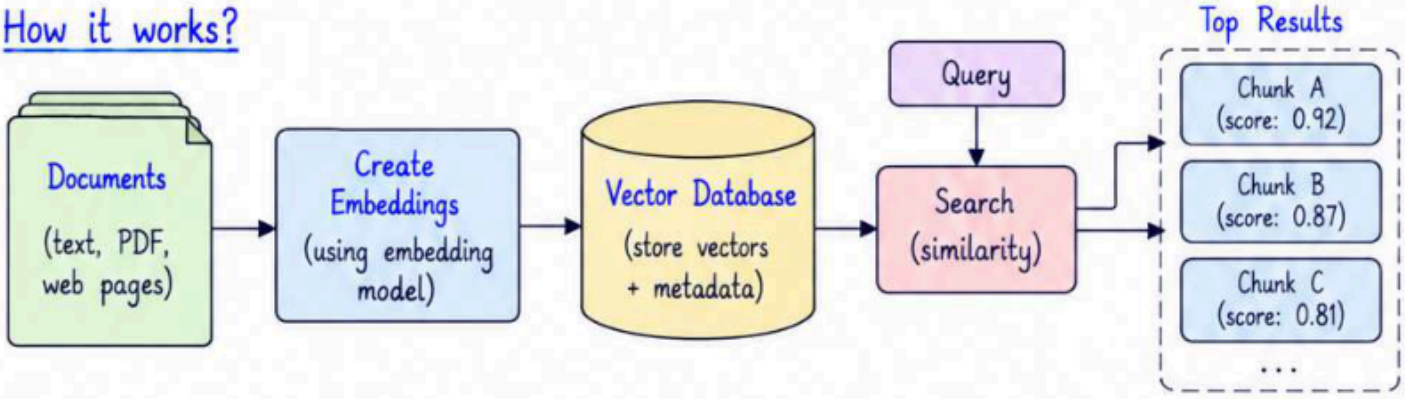
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8 Vector Database

Store and search embeddings

- A vector database stores embeddings (vectors) and allows fast similarity search.
- It helps find the most similar chunks/documents based on meaning.

How it works?



Key Points

- Stores high-dimensional vectors.
- Uses similarity search (e.g. cosine similarity).
- Stores metadata along with vectors.
- Optimized for fast search at scale.

Use Cases

- Retrieval in RAG systems
- Finding similar documents / content
- Chat with your data (PDFs, knowledge base)
- Recommendations, semantic search

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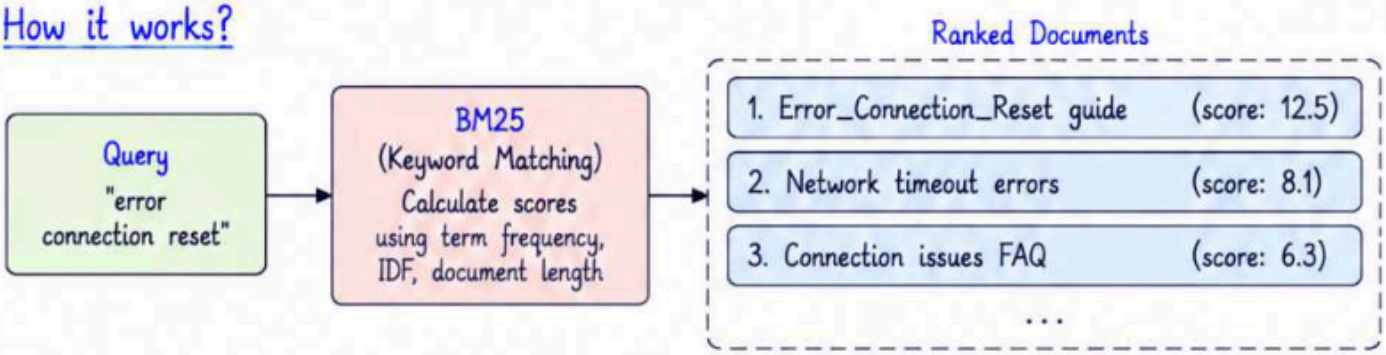
Key Ideas Behind Modern AI Systems

9 BM25

Keyword based search

- BM25 ranks documents based on keyword matching and term frequency (not meaning).
- It gives higher scores to documents where the query terms appear more often and are less common.

How it works?



Key Points

- Keyword based (exact term matching).
- Considers term frequency and document length.
- Fast and effective for exact terms (codes, names, errors).

Use Cases

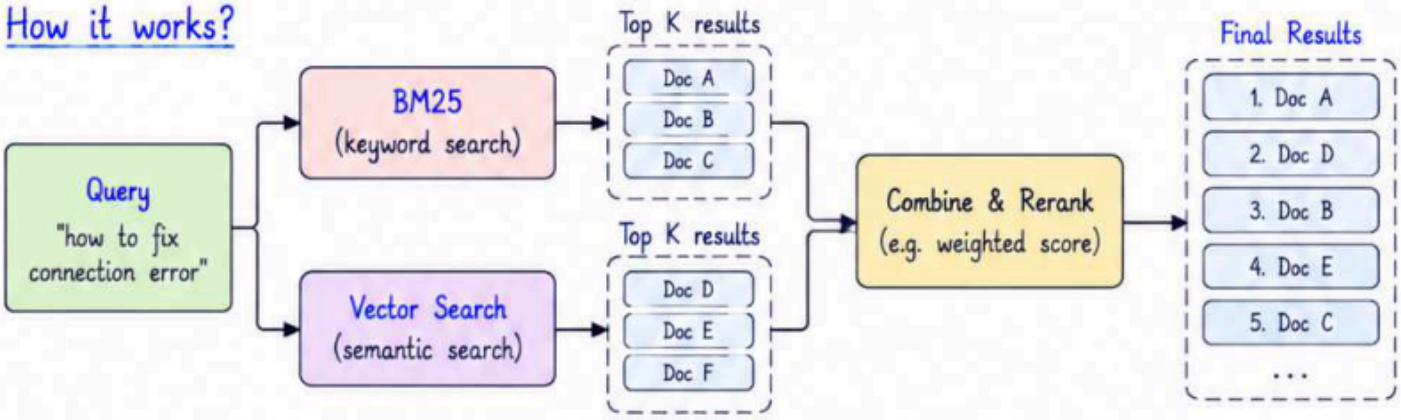
- Searching error codes and logs
- Technical documentation search
- Product names, exact keywords

10 Hybrid Search

Combine keyword and semantic search

- Hybrid search combines BM25 (keyword search) and vector search (semantic search).
- It merges results from both methods to get more relevant and accurate results.

How it works?



Key Points

- Combines keyword + semantic search.
- Better recall and relevance.
- Reduces chances of missing useful documents.

Use Cases

- RAG systems
- Searching technical and conceptual content
- Enterprise knowledge search
- Customer support / help centers

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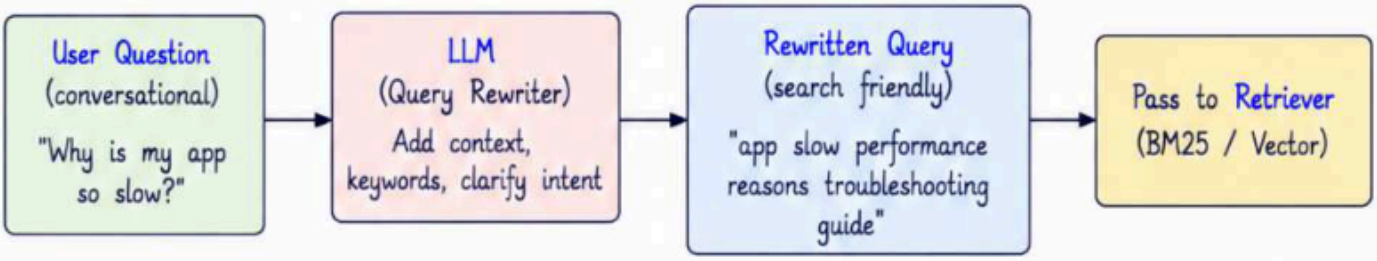
Key Ideas Behind Modern AI Systems

11 Query Rewriting

Turn conversation into a better search query

- Query rewriting converts a user's natural language question into a clear and explicit search query.
- It adds missing context, important keywords and keeps the original intent.

How it works?



Key Points

- Converts vague questions to clear search queries.
- Improves retrieval quality.
- Useful for conversational RAG.

Use Cases

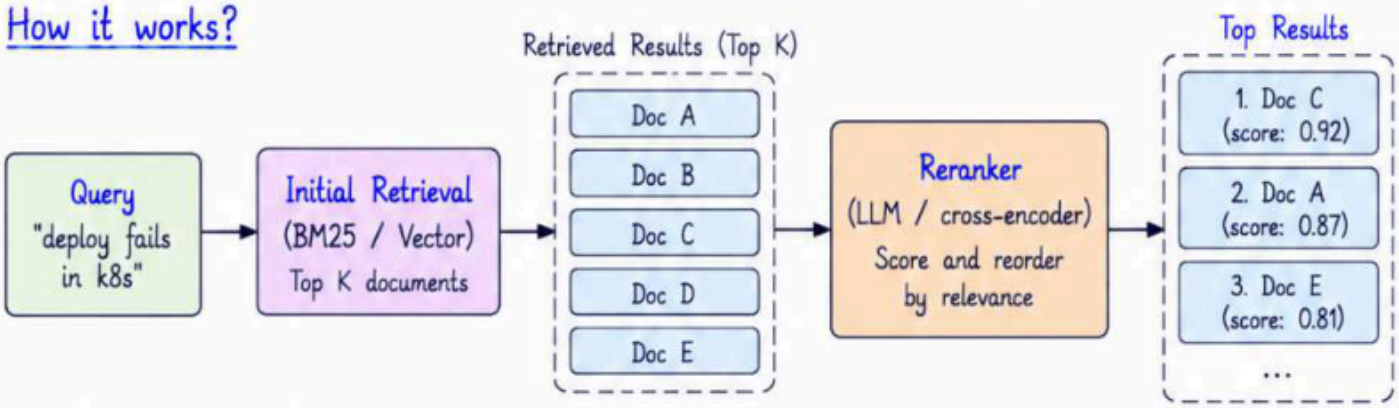
- Chatbots and assistant applications
- Technical support search
- Handling follow-up questions
- Enterprise knowledge bases

12 Reranking

Reorder candidates by relevance

- Reranking uses a more advanced model to carefully score and reorder the retrieved documents.
- It helps pick the most relevant results from the initial list.

How it works?



Key Points

- Uses a stronger model for better relevance.
- Considers full query and document meaning.
- Improves accuracy over initial retrieval.

Use Cases

- RAG systems
- Search and recommendation
- Filtering large number of results
- Enterprise and technical documentation

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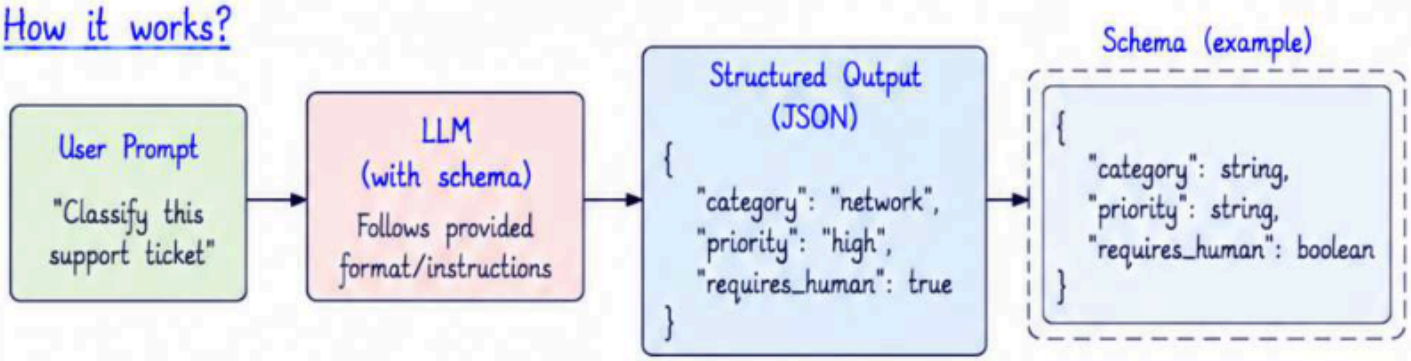
Key Ideas Behind Modern AI Systems

13 Structured Outputs

Make model responses follow a schema

- Structured outputs ensure the model returns data in a predefined format (e.g. JSON).
- This makes the response predictable and easier to consume in applications.

How it works?



Key Points

- Follows a defined schema or format.
- Reduces parsing errors.
- Useful for reliable integration with other systems.

Use Cases

- Auto-classifying support tickets
- Extracting structured data from documents
- Returning API-friendly responses
- Building reliable AI agents

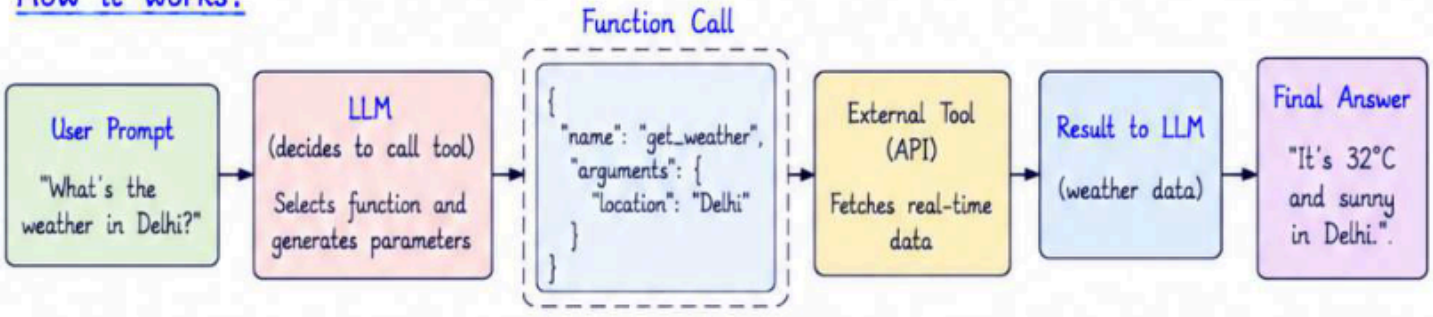
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14 Function Calling (Tool Calling)

Let the model use external tools

- Function calling allows the model to invoke external tools (APIs, databases, etc.) when needed.
- The model decides which tool to use and with what parameters.

How it works?



Key Points

- Enables the model to interact with real systems.
- Model chooses the right tool based on the query.
- Expands capabilities beyond training data.

Use Cases

- Calling APIs (weather, maps, payments)
- Querying databases (SQL)
- Automating real-world tasks
- Building AI agents and copilots

AI Engineering Concepts

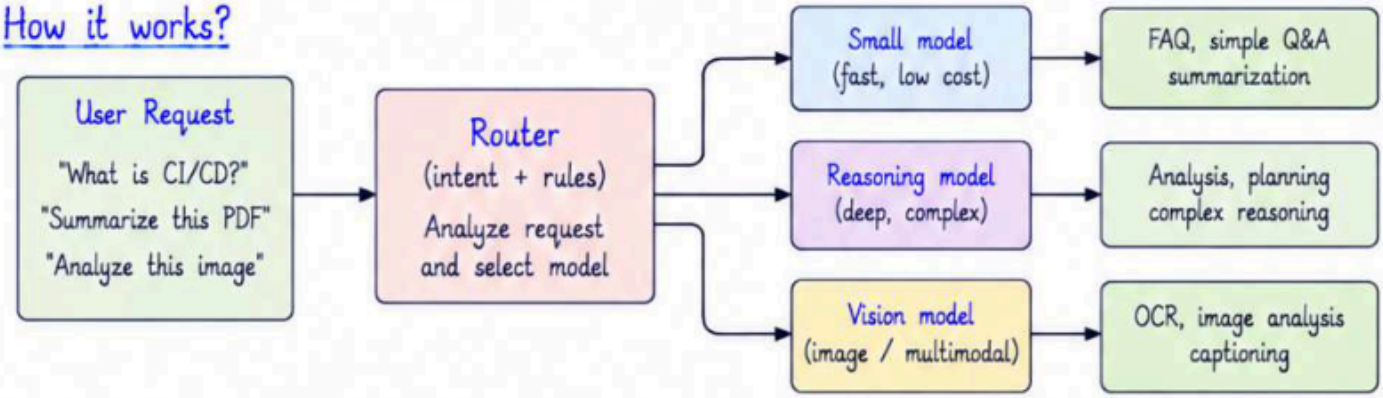
Key Ideas Behind Modern AI Systems

15 Model Routing

Pick the right model for each request

- Model routing analyzes the request and chooses the most suitable model based on intent, complexity, cost, and latency.
- It helps optimize performance and cost without always using the largest model.

How it works?



Key Points

- Routes requests to the right model.
- Balances cost, latency and quality.
- Can use rules, ML classifier or LLM itself.

Use Cases

- Customer support bots
- Multi-model applications (text, image, code)
- Cost optimization in production
- Handling different task types efficiently

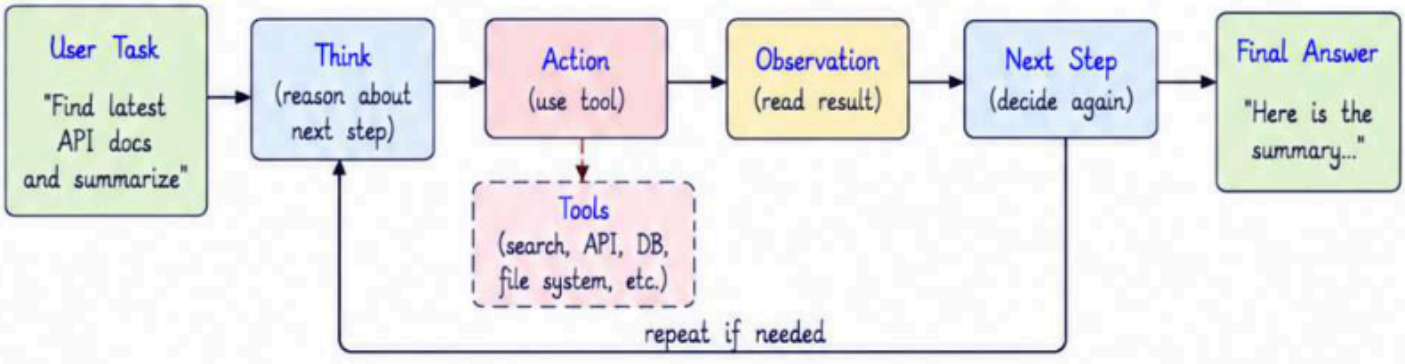
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16 ReAct (Agent Loop)

Reason, act, observe, then decide again

- ReAct combines reasoning and action by letting the model think, use tools, observe the results, and continue.
- It works in a loop until the task is complete.

How it works?



Key Points

- Interleaves reasoning and actions.
- Uses external tools to get real-time information.
- Continues in a loop until the goal is achieved.

Use Cases

- Research and information gathering
- Data analysis with tools (Python, SQL)
- Customer support automation
- Complex multi-step tasks

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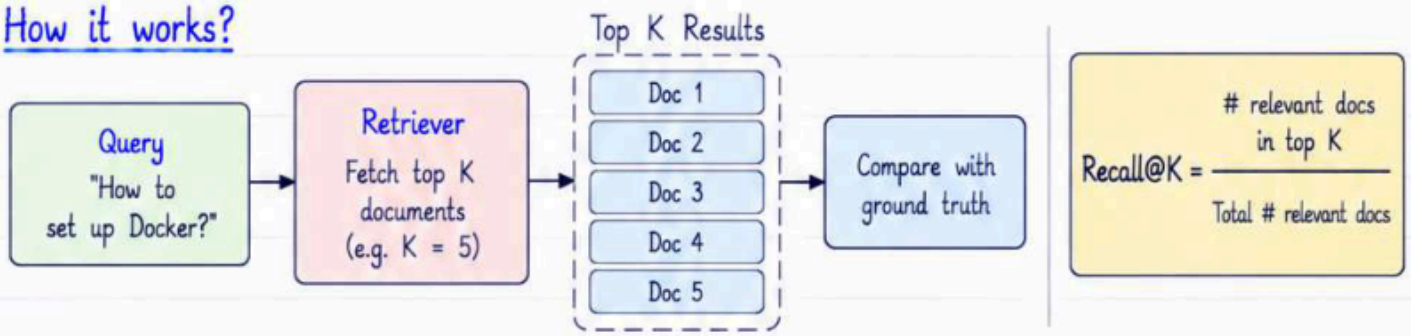
Key Ideas Behind Modern AI Systems

17 Recall@K

Measure how many relevant documents are retrieved

- Recall@K tells us what percentage of relevant documents are present in the top K results.
- It helps evaluate the retrieval quality in RAG systems.

How it works?



Key Points

- Higher recall means more relevant documents retrieved.
- Typically used with a predefined set of relevant documents (ground truth).
- Common values of K: 5, 10, 20.

Use Cases

- Evaluating RAG retrieval performance
- Comparing different embedding models
- Tuning chunking and search strategies
- Offline evaluation of knowledge bases

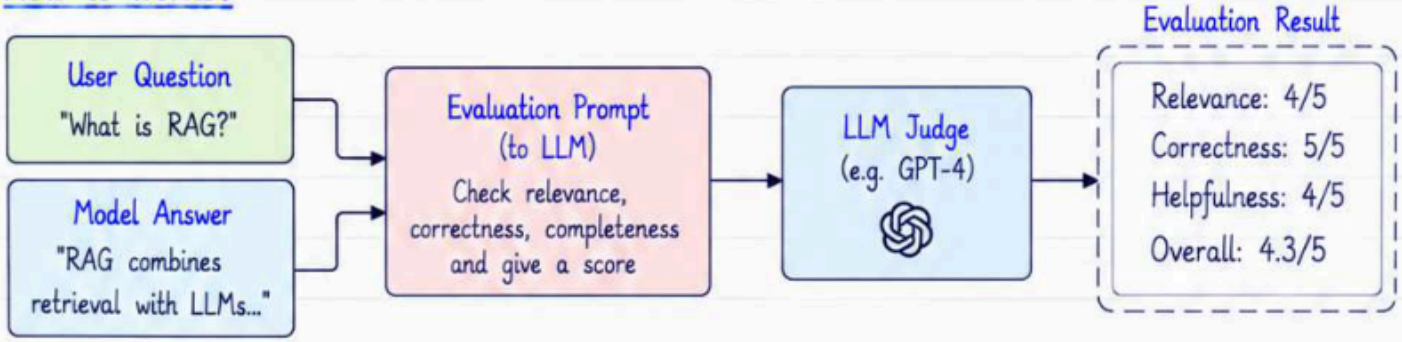
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18 LLM-as-a-Judge

Use LLMs to evaluate the quality of responses

- LLM-as-a-Judge uses a powerful LLM to assess the relevance, correctness and helpfulness of a response.
- It provides a scalable way to evaluate RAG outputs without extensive human annotation.

How it works?



Key Points

- Uses LLM to grade or compare responses.
- Can use a fixed rubric or scoring criteria.
- Useful for subjective and open-ended questions.

Use Cases

- Evaluating RAG answer quality
- Comparing different prompts / models
- Automated evaluation in CI/CD pipelines
- Reducing manual review effort

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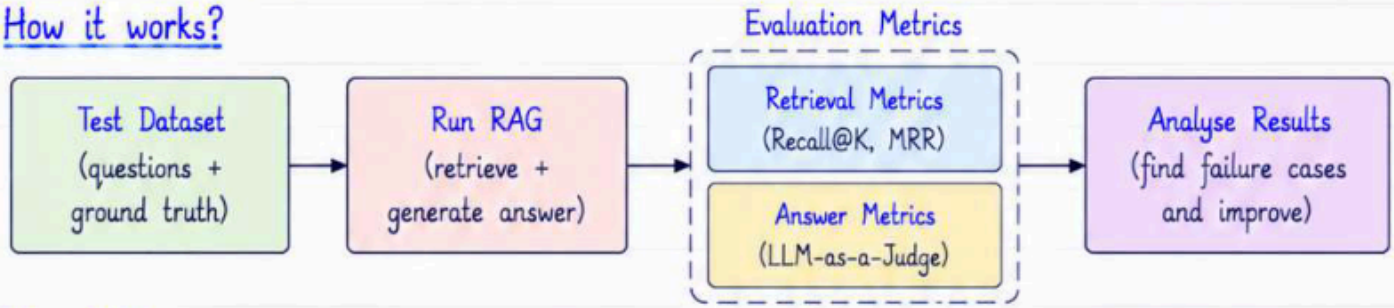
Key Ideas Behind Modern AI Systems

19 RAG Evaluation Pipeline

Evaluate the end-to-end RAG system

- A complete evaluation looks at both retrieval quality and answer quality.
- It helps identify gaps and continuously improve the system.

How it works?



Key Points

- Evaluate retrieval and generation separately.
- Use automated metrics + manual review.
- Track performance over time.
- Helps in continuous improvement.

Common Metrics

- Recall@K - retrieval coverage
- MRR - rank of first relevant document
- Faithfulness - answer grounded in context
- Correctness - factually accurate answer
- Helpfulness - useful and complete response

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20 Production Considerations & Best Practices

Make RAG systems reliable and scalable

- Moving from a prototype to production requires monitoring, safety and continuous improvements.
- Following best practices ensures better performance and user trust.

Key Areas

Monitoring
- Track latency
- Monitor retrieval and answer quality
- Log user feedback

Safety & Guardrails
- Use content filters
- Avoid hallucinations
- Set boundaries for sensitive topics

Scalability
- Use caching
- Optimize vector DB
- Handle high traffic

Continuous Improvement
- Use real user data
- Analyze failure cases
- Update data and prompts regularly

Cost Optimization
- Use model routing
- Cache responses
- Choose right model for each task

Best Practices

- Keep data up to date.
- Use hybrid search (keyword + vector).
- Evaluate regularly and track key metrics.
- Start simple, iterate based on real usage.

Real-world Use Cases (RAG)

- Enterprise document search
- Customer support assistants
- Internal knowledge copilots
- Research and analysis tools

Build → Evaluate → Improve → Scale

RAG turns information into useful answers!

Keep Learning
Keep Building
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